



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering

K. J. Somaiya School of Engineering, Mumbai-77

(A Constituent College of Somaiya Vidyavihar University)



Department of Electronics Engineering

Course Name:	Object Oriented Programming	Semester:	III
D a t e o f Performance:	___ / ___ / ____	Batch No:	Batch _A-4_
Faculty Name:	Prof.	Roll No:	16015024082
Faculty Sign & Date:		Grade/Marks:	___/15

Experiment No: 6
Title: Exception Handling in Java

Aim and Objective of the Experiment:

Learn the how to handle exceptions in Java

COs to be achieved:

CO3: Define exceptions and use I/O streams.

Tools used:

Theory:



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1. (About types of exceptions, the methods used for exceptions and keywords used for exception handling)

Code:

1. The program takes two numbers from users in the command prompt, Num1 and Num2. The division of Num1 and Num2 is displayed in the Result. If Num1 or Num2 were not an integer, the program would throw a NumberFormatException. If Num2 were Zero, the program would throw an Arithmetic Exception and Display the exception in the result.
 - a. Using try and multiple catches.
 - b. Using multiple exceptions in a single catch block.
2. Create a user defined exception subclass TimeException with necessary constructors and overridden toString method. Write a program which accepts two integers with time in minutes and seconds and find the sum. It throws an object of the TimeException class if the value exceeds 60seconds otherwise it displays the total time. On printing, the exception object should display an exception name, appropriate message for exception



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```
1.a
import java.util.Scanner;

public class DivisionExample {
    public static void main(String[] args) {
        Scanner sc = new Scanner(system.in);
        System.out.print("Enter first number: ");
        String n1 = sc.nextLine();
        System.out.print("Enter second number: ");
        String n2 = sc.nextLine();

        try {
            int num1 = Integer.parseInt(n1);
            int num2 = Integer.parseInt(n2);
            int result = num1 / num2;
            System.out.println("Result: " + result);
        } catch (NumberFormatException e) {
            System.out.println("Invalid input. Please enter integers only.");
        } catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero.");
        }
    }
}
```



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1.B

```
import java.util.Scanner;

public class DivisionSingleCatch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(system.in);
        System.out.print("Enter first number: ");
        String n1 = sc.nextLine();
        System.out.print("Enter second number: ");
        String n2 = sc.nextLine();

        try {
            int num1 = Integer.parseInt(n1);
            int num2 = Integer.parseInt(n2);
            int result = num1 / num2;
            System.out.println("Result: " + result);
        } catch (NumberFormatException | ArithmeticException e) {
            System.out.println("Exception occurred: " + e);
        }
    }
}
```



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```
2
class TimeException extends Exception {
    public TimeException(String msg) {
        super(msg);
    }
    public String toString() {
        return "TimeException: " + getMessage();
    }
}

public class TimeSum {
    public static void main(String[] args) {
        try {
            int m1 = 5, s1 = 40;
            int m2 = 2, s2 = 35;
            int totalM = m1 + m2;
            int totalS = s1 + s2;
            if (totalS > 60) throw new TimeException("Seconds exceeded 60");
            System.out.println("Total time: " + totalM + " min " + totalS + " sec");
        } catch (TimeException e) {
            System.out.println(e);
        }
    }
}
```

Output:



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Post Lab Subjective/Objective type Questions:



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1. Compare throw and throws.

->

1) throw is used inside a method to explicitly throw an exception object.

2) throws is used in a method declaration to specify that the method may throw certain exceptions to the caller.

2. What are the advantages of using exception handling?

- Separates error-handling code from normal logic.
- Allows program continuation after handling errors.
- Provides clear, maintainable and more robust code.

3. What is the use of finally block?

1) The finally block is used to execute important cleanup code (like closing files or releasing resources) regardless of whether an exception occurs or not.

4. Suppose the statement2 causes an exception in following try-catch block:

```
try {  
    statement1;  
    statement2;  
    statement3;  
}  
catch(Exception1 e1) {  
}  
catch(Exception2 e2){  
}  
  
statement4;
```

Answer the following questions:

- Will statement3 be executed?
- If the exception is not caught, will statement4 be executed?



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- If the exception is caught in the catch block, will statement4 be executed?
- If the exception is passed to the caller, will the statement4 be executed?

If statement2 causes an exception:

- statement3 will not be executed.
- If the exception is not caught, statement4 will not be executed.
- If the exception is caught, statement4 will be executed.
- If the exception is passed to the caller, statement4 will not be executed.

Conclusion:

In this experiment we learned how to use try, catch, finally, multiple-catch, and user-defined exceptions to handle runtime errors gracefully, improving program reliability and maintainability.

Signature of faculty in-charge with Date: